Beltrami PPE Swabs

Methodology

PPE swabs were run by RT-QuIC at 42°C for 72 hours. Maxpoint ratios and max slope values were used as predictors in a generalized linear model for the following factors:

1. Location (dump site, negative site, perimeter)
2. Timing (pre/post-sampling and post-decon)
3. Surface (the surface that was swabbed).

Least-square means were calculated, and pairwise comparisons were performed for each factor. Contrasts with a p-value ≤0.05 were considered significant.

Results

Table 1: Maxpoint ratio results. Significant contrasts are highlighted in red.

Factor	Contrast	Estimate	Std.error	p.value	Metric	Result
Location	negative site - perimeter	-1.198	0.3529	0.002121	MPR	**
Location	negative site - dump site	-0.768	0.3082	0.03462	MPR	*
Location	dump site - perimeter	-0.4296	0.2308	0.1509	MPR	ns
Timing	pre sampling - post decon	-1.655	0.3856	6.13e-05	MPR	***
Timing	post decon - post sampling	1.397	0.3823	0.0008206	MPR	***
Timing	pre sampling - post sampling	-0.2581	0.1854	0.3457	MPR	ns
Surface	tyvek - boot	-1.211	0.2268	7.867e-07	MPR	***
Surface	blank - boot	-1.522	0.304	4.389e-06	MPR	***
Surface	boot - plastic_boot	0.6559	0.2478	0.04144	MPR	*
Surface	blank - plastic_boot	-0.8658	0.3365	0.05042	MPR	ns
Surface	tyvek - plastic_boot	-0.5554	0.271	0.1713	MPR	ns
Surface	tyvek - blank	0.3104	0.3232	0.7719	MPR	ns

Table 2: Max slope results. Significant contrasts are highlighted in red.

Factor	Contrast	Estimate	Std.error	p.value	Metric	Result
Location	negative site - perimeter	-0.4147	0.1322	0.005111	MS	**
Location	negative site - dump site	-0.3111	0.1155	0.01987	MS	*
Location	dump site - perimeter	-0.1036	0.08648	0.455	MS	ns
Timing	pre sampling - post decon	-0.6268	0.1445	5.019e-05	MS	***
Timing	post decon - post sampling	0.5366	0.1433	0.0005797	MS	***
Timing	pre sampling - post sampling	-0.09028	0.06949	0.3962	MS	ns
Surface	tyvek - boot	-0.3705	0.08499	9.054e-05	MS	***
Surface	blank - boot	-0.4204	0.1139	0.001397	MS	**
Surface	boot - plastic_boot	0.199	0.09288	0.1411	MS	ns
Surface	blank - plastic_boot	-0.2214	0.1261	0.2958	MS	ns
Surface	tyvek - plastic_boot	-0.1715	0.1016	0.3306	MS	ns
Surface	tyvek - blank	0.04993	0.1211	0.9764	MS	ns

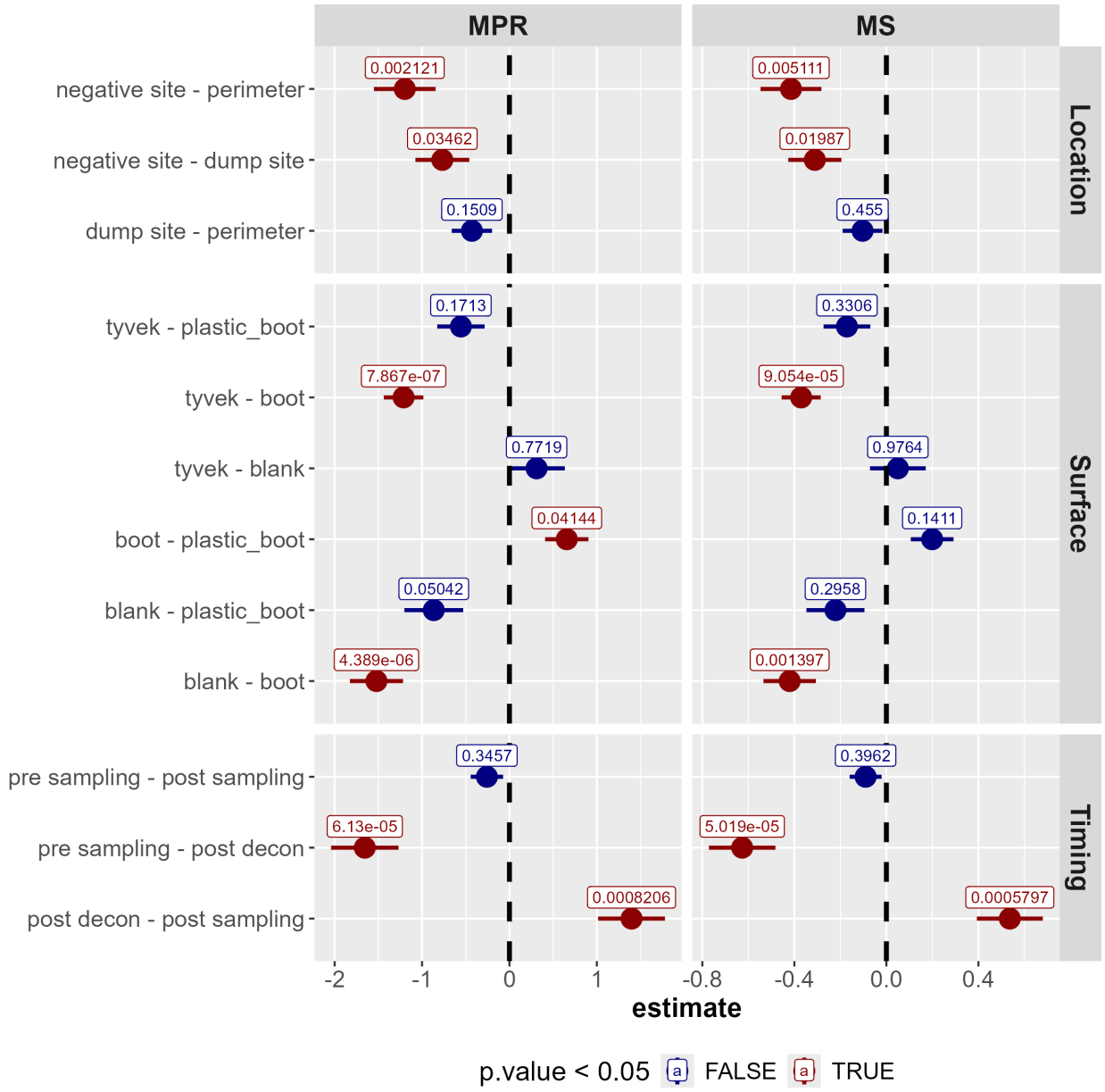


Figure 1: Estimates of each contrast. Plot is facetted by factor and metric (MPR or MS). Estimates are labelled with their p-value and colored based on significance.

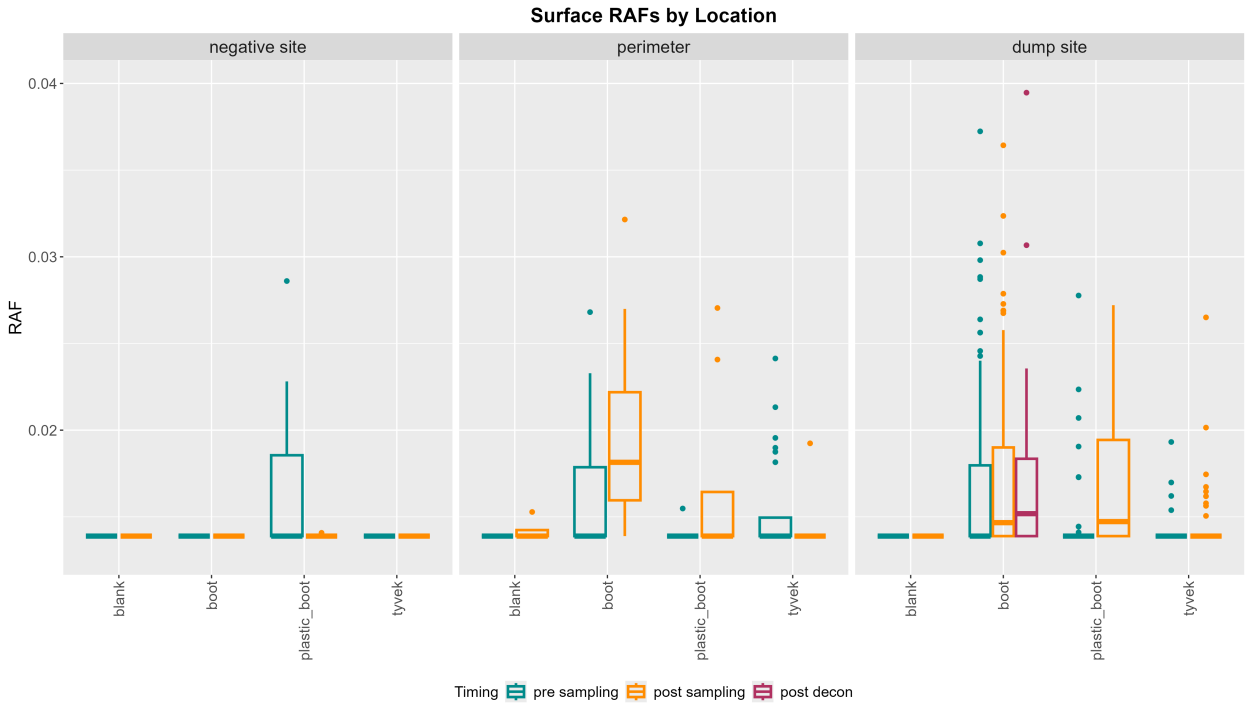


Figure 2: Rates of amyloid formation by swab surface. Figure is facetted based on location and colored based on timing.

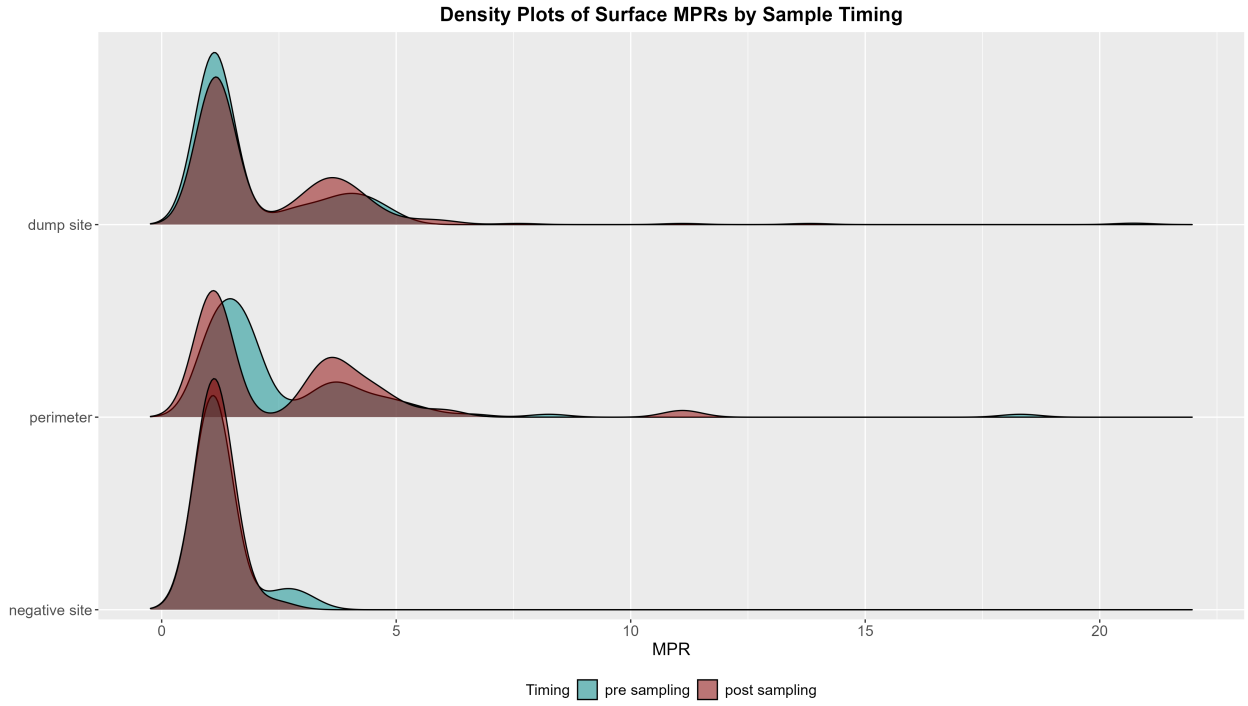


Figure 3: Density plot of MPRs for each location. Curves are filled based on timing.